

DATA VORTEX

Exploratory Data Analysis (EDA) & Data Cleaning Report

Theme: Rebuilding the Social Engine

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1. Executive Summary

The Social Engine Posts dataset contains post-level social media activity linked to user profiles and was analyzed to assess data quality, establish a reliable analytical dataset, and identify meaningful patterns in platform activity and user engagement. The original dataset contained 12,360 records across 8 columns and included several deliberate data-quality issues, such as missing values, exact duplicates, invalid engagement values, inconsistent text representations, encoding errors, HTML artifacts, and mixed timestamp formats.

A systematic Exploratory Data Analysis (EDA) and data-cleaning process was conducted to identify, validate, and resolve these issues while preserving the underlying information wherever possible. The analysis identified 1,846 missing platform values, 1,746 explicit missing text values, 1,858 missing likes, 525 negative likes, and 360 exact duplicate rows. Additional quality issues included 94 NULL-prefixed text values, HTML/HTML-entity artifacts, 287 encoding-corrupted text values, and multiple timestamp representations.

Data preparation reduced the dataset from 12,360 to 12,000 records through exact duplicate removal. Missing platform values were conservatively categorized as Unknown, while text fields were cleaned by removing NULL prefixes and HTML/entity artifacts. The 287 identified encoding-corrupted text values were repaired, and timestamps were standardized into a consistent format. Negative likes were treated as invalid observations and replaced using the median of valid like values (2,502). These decisions were designed to minimize information loss while ensuring that the resulting dataset was suitable for reliable statistical and exploratory analysis.

Following data cleaning, the final dataset contained 12,000 rows and 8 columns, with zero missing values, zero exact duplicate rows, zero duplicate post IDs, and no negative engagement values. The cleaned dataset therefore provides a consistent foundation for subsequent analysis of social media behavior and engagement.

The EDA revealed that platform activity was broadly balanced, with no single platform overwhelmingly dominating the dataset. Monthly posting volume remained relatively stable, indicating no major structural changes in activity over the observed period. Engagement variables such as likes, shares, and comments showed negligible linear correlations, suggesting that high performance in one engagement metric does not necessarily imply proportionally high performance in another. Similarly, follower count showed negligible linear association with engagement, indicating that having more followers alone does not strongly determine post-level engagement.

Language representation was also relatively balanced across the dataset. When comparing high-performing posts, Facebook recorded the highest proportion of posts within the overall top 10% of likes at 10.99%, while YouTube recorded the lowest at 9.17%. The relatively narrow difference between platforms suggests that there is no overwhelming platform-level advantage in producing highly liked posts within this dataset.

Overall, the EDA demonstrates that careful data-quality treatment was essential before interpreting engagement patterns. The cleaned dataset supports reliable analysis of platform performance, engagement behavior, temporal trends, user activity, follower relationships, and language differences, while the exploratory findings indicate that social media engagement is influenced by factors beyond simple platform choice or follower count.

2. Objectives and Analytical Questions

The Exploratory Data Analysis (EDA) was conducted to systematically assess the structure, quality, and reliability of the social media dataset and to uncover meaningful patterns in user activity and engagement. The analysis focuses on engagement metrics, user characteristics, platform and language differences, and temporal trends. The findings are intended to provide an evidence-based understanding of the dataset while avoiding unsupported assumptions or causal interpretations.

Objectives:

Objective 1: Assess Data Quality and Structural Integrity

Evaluate the structure, completeness, consistency, and overall reliability of the Posts and Users datasets, including their variables, data types, and relationships.

Objective 2: Identify and Address Data Quality Issues

Detect and appropriately handle missing values, duplicate records, invalid or anomalous engagement values, inconsistent timestamp formats, and corrupted or unreliable text fields.

Objective 3: Analyze Engagement Distributions and Relationships

Examine the distributions of likes, shares, comments, and follower counts, and investigate relationships among these variables along with platform and language characteristics.

Objective 4: Identify Meaningful Patterns and Segment-Level Differences

Explore variations in engagement and user activity across platforms, language groups, users, and time periods, while distinguishing observable associations from unsupported causal claims

Key Analytical Questions

Engagement Distribution:

How are likes, shares, and comments distributed across posts, and are there any extreme or unusual observations that require further investigation?

Variable Relationships:

What relationships exist among likes, shares, comments, and follower count, and what do these relationships indicate about engagement patterns?

Platform and Language Segmentation:

How does engagement vary across different social media platforms and language groups, and are certain segments noticeably different from others?

Temporal Patterns:

How do posting volume and average engagement vary over the observed monthly period, and are there any notable temporal patterns or fluctuations?

Data Reliability:

To what extent do missing, duplicated, inconsistent, or anomalous records affect the reliability and interpretation of the observed patterns?

3. Dataset Overview

3.1 Dataset dimensions

Item	Before cleaning	After cleaning
Rows / observations	12,360	12,000
Columns / features	8	9
Dataset role	Corrupted Posts dataset	Cleaned Posts dataset
Users reference	1,500 rows × 5 columns	Validated; used for enrichment/EDA

The Posts dataset represents individual social media posts and their engagement counts, linked to a Users reference table through user_id. The cleaning target was Dataset 1 (Posts); the Users file was audited and used as a reference/enrichment source.

3.2 Variable description / Data dictionary

Variable	Type	Description	Role / cleaning note
post_id	Identifier	Unique identifier for a post.	Used to detect duplicate post IDs; retained unchanged.
user_id	Identifier	Identifier linking a post to a user profile.	Referential integrity checked; retained unchanged.
platform	Categorical	Platform associated with the post.	Missing values are represented as Unknown; categories are checked for consistency.
text_content	Text	Content of the social media post.	NULL, HTML/entity, whitespace, and encoding artifacts cleaned.
timestamp	Datetime	Date/time associated with the post.	Mixed source formats standardized to datetime.
likes	Integer count	Number of likes associated with a post.	Negative values are treated as invalid; missing/invalid values are median-imputed.
shares	Integer count	Number of shares associated with a post.	Complete and non-negative; no imputation required.
comments	Integer count	Number of comments associated with a post.	Complete and non-negative; no imputation required.
user_id (Users)	Identifier	User key in reference table.	1,500 unique IDs matched to all posts users.
location (Users)	Categorical/text	User location.	Audited; no substantive transformation.
language (Users)	Categorical	Language associated with the user.	Used for language-level EDA.
account_created (Users)	Datetime	Date the user account was created.	Converted from YYYY-MM-DD text to datetime.
follower_count (Users)	Integer count	Number of followers associated with a user.	Validated as non-negative; used for correlation analysis.

4. Initial Data Inspection

Before performing any data transformation or cleaning, a baseline audit of the raw datasets was conducted. The purpose of this inspection was to understand the original data structure, assess its quality, identify potential inconsistencies, and establish an audit trail for subsequent cleaning decisions.

4.1 Initial Inspection of the Posts Dataset

The following aspects of the raw Posts dataset were examined:

- Record structure: First and last observations were inspected to understand the format and content of the raw records.

- Dataset dimensions: The number of observations and variables was verified using the dataset shape.
- Data types and completeness: Column data types and non-null counts were examined to identify type inconsistencies and missing values.
- Descriptive statistics: Numerical variables, particularly likes, shares, and comments, were summarized to identify their ranges, central tendencies, and potential anomalies.
- Uniqueness and duplication: Unique-value counts and duplicate records were checked, including duplicate `post_id` values.
- Missing and invalid values: Missing values and potentially invalid engagement counts, such as negative values, were identified.
- Text quality: `text_content` was inspected for NULL values, placeholder text, HTML/entity artifacts, excessive whitespace, and encoding issues.
- Timestamp consistency: Timestamp values were examined for inconsistent formats and values that could not be reliably interpreted as dates.

4.2 Initial Inspection of the Users Dataset

The Users reference dataset was separately audited to assess its suitability for enrichment and relational analysis. The inspection covered:

- Uniqueness and completeness of `user_id`.
- Missing values across user attributes.
- Validity and distribution of follower counts.
- Distribution of users across language categories.
- Consistency of `account_created` date values.
- Referential consistency between `user_id` values in the Posts and Users datasets.

4.3 Inspection Methods

The baseline audit was supported by standard exploratory data analysis operations, including:

`head()`, `tail()`, `shape`, `info()`, `describe()`, `dtypes`, `nunique()`, `isna().sum()`, and `duplicated().sum()`.

These checks established the pre-cleaning state of the datasets and provided the evidence required to determine which observations and variables required transformation, correction, removal, or retention in the subsequent data-cleaning stage

5. Data Types

Data types were reviewed because analytical operations depend on semantic types. Identifiers should remain identifiers, categorical variables should have consistent labels, count variables should be numeric, and temporal variables should be datetime objects.

Variable	Initial / observed form	Final form	Why
post_id	object	object	Identifier: numeric conversion would add no analytical value.
user_id	object	object	Identifier: preserved as text to avoid treating IDs as quantities.
platform	object/categorical text	object with 6 canonical labels	Consistent grouping and explicit unknown category.
text_content	object	cleaned text	Required removal of artifacts while preserving content.
timestamp	mixed strings/numeric representations	datetime64[ns]	Required for chronological and temporal analysis.
likes	float after imputation	int64	Engagement count is discrete; integer storage reflects semantics.
shares	int64	int64	Already appropriate for a count.
comments	int64	int64	Already appropriate for a count.
account_created	object	datetime64[ns]	Required for valid date operations; all dates followed YYYY-MM-DD.

The timestamp conversion was necessary because the source contained 10-digit Unix timestamps, DD-MM-YYYY date strings, and ISO-like timestamps. A single datetime representation permits sorting, date-range validation, month extraction, and time-based aggregation.

6. Missing Value Analysis

Missingness was quantified before any cleaning or imputation to establish a reliable baseline and ensure that each subsequent treatment decision could be justified using the observed data.

Column	Missing count	Missing %
post_id	0	0.00%
user_id	0	0.00%
platform	1846	14.94%
text_content	1746	14.13%
timestamp	0	0.00%
likes	1858	15.03%
shares	0	0.00%
comments	0	0.00%

The five user-level columns were complete, with no missing values. Within the Posts dataset, missingness was concentrated in platform, text_content, and likes, while post_id, user_id, timestamp, shares, and comments were complete.

Treatment of Missing Values

- Platform: All 1,846 missing platform values were assigned the category Unknown. Platform could not be inferred reliably from the available information, so no unsupported assumptions were introduced.

- Text Content: Missing text and NULL-prefixed text were treated as unavailable content rather than attempting to reconstruct or fabricate text. After accounting for these cases, 1,779 records were represented as Unknown in the cleaned dataset.
- Likes: Missing likes and invalid negative likes were handled as part of the same engagement-cleaning process after duplicate removal. Invalid negative values were excluded from the calculation of the replacement statistic. Missing/invalid values were then imputed using the median of valid non-negative likes, 2,502. The median was selected because it is less sensitive to extreme engagement values than the mean and provides a deterministic replacement.
- Shares and Comments: No imputation was required because both columns were complete and contained valid non-negative values.

Assumption: Unknown is used strictly as a missing-information marker. It does not imply that the actual platform or text content was literally recorded as "Unknown." This approach preserves the affected observations

7. Duplicate Analysis

The raw Posts dataset contained 360 exact duplicate rows. A corresponding 360 duplicate post_id values were identified, indicating that the duplicated records represented repeated copies of existing posts rather than distinct observations.

Exact duplicate rows were removed during the cleaning stage because retaining them would artificially inflate post counts, user activity, platform frequencies, monthly posting volume, and engagement statistics. Since post_id serves as the natural identifier for each post, no additional business-key deduplication was applied. This avoided the risk of incorrectly removing legitimate records without sufficient evidence that multiple distinct events could share the same post_id.

After deduplication, the Posts dataset contained 12,000 unique observations, compared with 12,360 observations before cleaning.

The Users dataset contained no exact duplicate rows and no duplicate user_id values. Therefore, no deduplication was required for the Users reference table.

Duplicate-checking methods:

df.duplicated().sum() was used to identify exact duplicate rows, while duplicate identifiers were assessed using the post_id and user_id fields.

8. Invalid & Inconsistent Data

The dataset was examined for values that were numerically invalid, categorically inconsistent, or textually corrupted. Each issue was investigated using both logical validation and diagnostic analysis before a treatment was applied.

8.1 Negative likes

The raw likes field contained 525 negative values, with a minimum of -4,987. Since likes represent a non-negative engagement count, negative values are semantically invalid.

The negative values were not simply multiplied by -1 , as doing so would assume that the corruption mechanism was a sign reversal without sufficient evidence to support that assumption.

Diagnostic analysis showed that the negative values were distributed across different platforms and users rather than being concentrated in a specific subgroup. Their absolute magnitudes also showed only negligible linear relationships with other engagement variables, including shares ($r \approx 0.023$) and comments ($r \approx -0.061$). Therefore, there was insufficient evidence that the negative values represented valid positive engagement counts.

The negative likes were consequently reclassified as missing values and handled using the same robust imputation strategy as the originally missing likes. The median of valid non-negative likes (2,502) was used for deterministic imputation.

8.2 Platform consistency

Platform labels were standardized and validated to prevent category fragmentation during aggregation and comparison.

The final set of platform categories was:

Facebook, YouTube, Twitter, Reddit, Instagram, and Unknown.

No lower/upper-case inconsistencies, leading/trailing whitespace variants, or duplicate category representations remained after cleaning. Missing platform values were retained as Unknown because the available information was insufficient to reliably infer the original platform.

8.3 Text inconsistencies and corruption

The `text_content` field contained several forms of textual inconsistency. The source data included 94 NULL-prefixed values and 1,004 rows containing HTML or HTML-entity patterns.

The following cleaning operations were applied:

- HTML tags were removed.
- HTML entities were decoded.
- Whitespace was normalized.
- NULL-prefixed values were treated as unavailable text rather than valid content.

A separate encoding audit identified 287 rows containing suspicious mojibake patterns, including malformed character sequences such as `Ã`, `Â`, and `â`. A controlled encoding-repair procedure was applied only to the affected records.

Examples such as `TravelÃ©` $\rightarrow$ `Travelé` and `feedbackÃ©` $\rightarrow$ `feedbacké` confirmed that the repair restored the intended characters. A post-repair validation found zero remaining suspicious encoding artifacts, indicating that the encoding-cleaning step was successfully completed without unnecessarily modifying unaffected text.

9. Outlier Analysis

Outlier analysis was treated as an investigative step rather than an automatic deletion rule. Since likes, shares, and comments are bounded count variables, unusually large values may represent genuinely successful posts rather than data-entry errors. Therefore, statistical extremeness alone was not considered sufficient evidence for removal.

Metric	Min	Median	Q1	Q3	Max	Upper-bound observations
Likes	0	2502	1547	3450	5000	4 exactly at 5,000 (0.03%)
Shares	0	1018	510	1501	2000	3 exactly at 2,000 (0.02%)
Comments	0	503	253	755	1000	10 exactly at 1,000 (0.08%)

The upper-bound observations were extremely rare, accounting for less than 0.1% of records for each engagement metric. Furthermore, the values occurred exactly at the defined upper limits rather than appearing as isolated values beyond the permissible range.

These observations were therefore retained because there was no evidence that they represented erroneous records. Removing them solely because they were statistically extreme could eliminate legitimate high-engagement posts.

The analysis follows the principle that a statistical outlier is not automatically a data error. Extreme observations were retained unless there was independent evidence of invalidity.

10. Distribution Analysis

Distribution analysis was conducted to understand the central tendency, spread, range, and concentration of the three primary engagement metrics: likes, shares, and comments

Metric	N	Mean	Std	Min	Q1	Median	Q3	P95	P99	Max
likes	12000	2495.15	1291.58	0	1547.00	2502.00	3450.00	4685.00	4937.02	5000
shares	12000	1007.17	575.07	0	510.00	1018.00	1501.00	1901.00	1977.01	2000
comments	12000	504.35	288.68	0	253.00	503.00	755.00	948.00	989.00	1000

Likes have the widest absolute range (0–5,000), followed by shares (0–2,000) and comments (0–1,000). The three variables show substantial dispersion, as indicated by their standard deviations and the separation between their lower and upper percentiles.

For likes, the mean (2,495.15) and median (2,502) are almost identical, indicating that the distribution is broadly centered around 2,500. Shares show a similar pattern, with a mean of 1,007.17 and median of 1,018, while comments have a mean of 504.35 and median of 503.

The high percentiles demonstrate that substantial engagement extends across a meaningful portion of the upper tail. For example, 99% of likes are at or below 4,937, compared with the maximum of 5,000. Similar patterns are observed for shares and comments.

However, the interpretation of the likes distribution should account for the median-imputation step used during data cleaning. Imputed values can increase the concentration of observations around the median and may therefore affect measures of central tendency and the apparent shape of the distribution.

11. Categorical Variable Analysis

Categorical variables were analyzed to assess the composition of the dataset and determine whether any platform or language category was disproportionately represented. This analysis focused on platform and language distributions.

Platform distribution:

Platform	Posts	Percentage
Facebook	2074	17.28%
YouTube	2073	17.27%
Twitter	2049	17.08%
Reddit	2031	16.92%
Instagram	1989	16.58%
Unknown	1784	14.87%

The five identified platforms show a broadly balanced distribution, with their individual shares ranging from 16.58% to 17.28%. The difference between the highest and lowest identified platform is only 0.70 percentage points, indicating that no single known platform dominates the dataset.

The Unknown category accounts for 1,784 posts (14.87%). These observations were retained rather than excluded because the original platform information was unavailable and could not be reliably inferred. Keeping Unknown as an explicit category preserves these records and makes the extent of missing platform information transparent.

Finding: Platform composition is sufficiently balanced to support meaningful descriptive comparisons across the five identified platforms, while the relatively large Unknown group should be considered when interpreting platform-level results.

Language distribution:

Language	Posts	Percentage
zh	1369	11.41%
ja	1298	10.82%
hi	1291	10.76%
en	1220	10.17%
fr	1156	9.63%
de	1149	9.57%
es	1143	9.52%
pt	1132	9.43%
ar	1123	9.36%
ru	1119	9.32%

Language representation is also relatively balanced. Individual languages account for between 9.32% and 11.41% of posts, producing a maximum difference of only 2.09 percentage points between the most and least represented languages.

Chinese (zh) is the most represented language at 11.41%, while Russian (ru) is the least represented at 9.32%. No language accounts for a disproportionately large share of observations.

The relatively narrow range of language frequencies indicates that the dataset does not heavily favor any single language. This provides a reasonable basis for descriptive comparisons of engagement across language groups, while recognizing that frequency balance alone does not establish that engagement behavior is statistically equivalent across languages.

12. Relationship Analysis

Relationship analysis was performed to examine whether engagement measures were linearly associated with one another and whether engagement varied systematically across platforms, follower counts, and languages. Pearson correlation was used for numerical variables, while group-level averages were used for categorical comparisons. These analyses are descriptive and do not establish causation

12.1 Numerical ↔ Numerical: engagement correlation

	Likes	Shares	Comments
Likes	1.000	-0.003	0.012
Shares	-0.003	1.000	0.024
Comments	0.012	0.024	1.000

The Pearson correlations between the engagement measures are all extremely close to zero:

- Likes ↔ Shares: -0.003
- Likes ↔ Comments: 0.012
- Shares ↔ Comments: 0.024

These results indicate negligible linear association among likes, shares, and comments in the analyzed dataset. In other words, a post receiving a high number of likes does not necessarily receive proportionally high shares or comments.

This is a descriptive Pearson-correlation finding only. Near-zero correlation does not rule out nonlinear relationships, interactions, or other forms of association, and it should not be interpreted as evidence of causation.

12.2 Platform ↔ Engagement

Platform	Avg likes	Avg shares	Avg comments
Facebook	2521.94	984.17	506.94
Instagram	2495.99	1040.84	499.80
Reddit	2490.37	1002.23	511.18
Twitter	2452.51	1005.39	506.13
Unknown	2484.95	998.61	496.54
YouTube	2523.14	1011.83	504.38

Average engagement varies only modestly across platforms.

- YouTube has the highest average likes (2,523.14), closely followed by Facebook (2,521.94).
- Instagram has the highest average shares (1,040.84).
- Reddit has the highest average comments (511.18).

The differences between platform averages are relatively small compared with the overall engagement ranges. Therefore, the data do not provide evidence that one platform overwhelmingly dominates engagement.

These are group-level descriptive comparisons and should not be interpreted as causal effects of platform choice.

12.3 Follower count ↔ Engagement

Following a complete merge with the user-level data, follower_count contained no missing values. Pearson correlations between follower count and engagement were:

- Follower count ↔ Likes: 0.008
- Follower count ↔ Shares: -0.014
- Follower count ↔ Comments: -0.000

All three correlations are effectively zero, indicating negligible linear association between follower count and the three engagement measures.

Thus, within this dataset, having more followers does not appear to be a meaningful linear predictor of post-level likes, shares, or comments.

However, this result does not rule out nonlinear relationships, interaction effects, or differences across individual users or other segments.

12.4 Language ↔ Engagement

Language	Avg likes	Avg shares	Avg comments
en	2556.70	997.11	504.18
ar	2536.54	1032.73	495.48
ja	2508.02	995.97	510.10
zh	2505.29	1022.98	500.12
fr	2499.38	980.25	519.76
pt	2493.83	1036.90	509.70
ru	2488.13	991.12	512.72
de	2477.94	1003.82	498.80
es	2467.65	1032.32	488.74
hi	2420.40	981.56	503.90

Among the listed language groups:

- English has the highest average likes (2,556.70).
- Hindi has the lowest average likes (2,420.40).
- Portuguese has the highest average shares (1,036.90).
- French has the highest average comments (519.76).

Although differences exist between language groups, they are descriptive differences in group averages and should not be interpreted as evidence that language itself causes higher or lower engagement.

13. Time-Series / Temporal Analysis

Temporal analysis was conducted to examine posting volume and average engagement over time and to determine whether the dataset exhibited sustained trends or meaningful seasonal changes.

The standardized timestamp range extends from 2024-05-01 00:00:00 to 2025-04-30 21:57:10, covering approximately one year of activity.

Month	Posts	Avg likes	Avg shares	Avg comments
2024-05	1038	2463.67	992.53	510.56
2024-06	979	2429.34	998.87	504.01
2024-07	996	2463.60	1025.69	485.13
2024-08	1015	2568.97	1019.43	494.40
2024-09	974	2530.08	1002.43	492.42
2024-10	1029	2539.16	1004.93	511.66
2024-11	1025	2509.93	1000.55	514.73
2024-12	1035	2510.14	1039.64	504.36
2025-01	1007	2429.86	997.88	499.66
2025-02	914	2461.46	999.89	510.42
2025-03	1013	2547.34	1002.80	509.87
2025-04	975	2482.12	999.93	514.71

Monthly posting volume ranges from 914 posts in February 2025 to 1,038 posts in May 2024. The relatively narrow range indicates that posting activity remained broadly stable throughout the observation period.

Average likes fluctuate between approximately 2,429 and 2,569, while average shares remain close to 1,000 per month and average comments remain around 500 per month. No sustained upward or downward movement is apparent across the twelve-month period.

The temporary increases or decreases observed in individual months therefore appear better characterized as local month-to-month variation rather than a persistent temporal trend.

The dataset exhibits relatively stable posting volume and engagement levels over time, with no strong evidence of a sustained trend or major structural shift during the observed period.

14. Feature Engineering

Feature engineering was performed only where an additional variable was required to support the exploratory analysis. A temporary month variable was derived from the standardized timestamp field to enable monthly aggregation and temporal trend analysis.

```
clean["month"] = clean["timestamp"].dt.to_period("M")
```

The month variable was used as an analytical helper variable for monthly post-volume and engagement analysis. It was intentionally removed before final export so that the submitted Posts CSV retained the original 8 dataset columns.

Deriving month from the standardized timestamp ensures consistent monthly grouping without altering the original timestamp information. Since it was created solely to support EDA, retaining it in the final dataset was unnecessary.

15. Data Transformation Summary

Issue / observation	Action	Reason	Validation / result
360 exact duplicate rows	Removed	Prevent double-counting identical observations	12,360 → 12,000; duplicates = 0
1,846 missing platform values	Filled with Unknown	Avoid unsupported platform inference	6 consistent categories; no missing
94 NULL-prefixed text values	Treated as missing	NULL is a placeholder, not content	Missing text represented as Unknown
1,004 HTML/entity rows	Removed tags + decoded entities	Remove formatting artifacts while preserving text	No remaining HTML/entity artifacts
287 encoding-corrupted text rows	Targeted encoding repair	Restore demonstrably recoverable characters	0 suspicious encoding artifacts
525 negative likes	Set invalid negatives to missing; median imputation	Negative count is invalid; avoid assuming sign reversal	Negative likes = 0; missing likes = 0
Mixed timestamp representations	Parsed to datetime	Enable valid temporal analysis	Invalid timestamps = 0
account_created as text	Converted to datetime	Enable date operations and consistency	0 missing; range 2023-01-01 to 2023-12-31
likes as float after imputation	Converted to int64	Count variable is discrete	Final likes dtype int64
EDA month helper	Removed before export	Not an original source field	Final CSV has original 8 columns

16. Before vs After Comparison

The effectiveness of the data-cleaning process was assessed by comparing key data-quality indicators before and after cleaning.

Quality dimension	Before cleaning	After cleaning
Rows	12,360	12,000
Columns	8	8
Missing values in Posts	5,450 across affected fields	0
Exact duplicate rows	360	0
Duplicate post IDs	360	0
Negative likes	525	0
Invalid timestamps	0 after parsing	0
HTML/entity artifacts	1,004 rows detected	0 remaining
Suspicious encoding artifacts	287 detected after initial text cleanup	0 remaining
Platform labels	Missing values present	6 explicit categories including Unknown

The comparison demonstrates that the cleaning process resolved the identified structural, numerical, textual, and missing-value issues while preserving the original 8-column schema.

The reduction from 12,360 to 12,000 rows resulted from removing 360 exact duplicate records. No additional business-key deduplication was imposed because post_id provides the natural record identifier and the duplicate observations were already exact copies.

The final state is not simply "better" because it contains fewer missing values. Rather, each transformation followed a documented rule: missing metadata and unavailable text were represented

explicitly as Unknown, invalid negative likes were treated as missing before imputation, and textual/encoding artifacts were repaired only where evidence supported the correction.

17. Key Findings

The EDA produced the following key findings:

- Platform activity is broadly balanced. The five identified platforms each account for approximately 16.6%–17.3% of posts, while Unknown represents 14.87% of records.
- Platform-level engagement differences are modest. No platform overwhelmingly dominates all three engagement measures. YouTube has the highest average likes (2,523.14), Instagram has the highest average shares (1,040.84), and Reddit has the highest average comments (511.18).
- Engagement dimensions have negligible linear relationships. Pearson correlations are approximately zero: likes–shares -0.003 , likes–comments 0.012 , and shares–comments 0.024 .
- High-like performance is similar across platforms. Facebook has the highest within-platform proportion of posts falling in the overall top 10% of likes (10.99%), while YouTube has the lowest (9.17%). The narrow 1.82-percentage-point spread does not indicate an overwhelming platform advantage.
- Posting activity is stable over time. Monthly post volume ranges from 914 to 1,038 posts across the 12-month observation period, with no sustained upward or downward trend.
- Follower count has negligible linear association with engagement. Correlations with likes (0.008), shares (-0.014), and comments (-0.000) are effectively zero.
- Language groups are reasonably balanced. Language representation ranges from 9.32% to 11.41% per language. English has the highest average likes, Portuguese the highest average shares, and French the highest average comments; these are descriptive group differences and do not establish causal effects.
- All Posts users were successfully matched to the Users reference table. The merged dataset therefore provided complete follower-count information for the relationship analysis. The average posting frequency was 8 posts per user, the median was 8, and 424 users had at least 10 posts.
- Extreme engagement values were retained. Rare observations at the upper bounds were not removed because there was no independent evidence that they represented erroneous data.

The cleaned dataset provides a consistent basis for exploratory analysis. The results suggest that engagement varies considerably between individual posts, while simple linear relationships between engagement metrics, follower count, platform, and language are generally weak or modest.

18. Explicit Assumptions

The following assumptions were applied during data cleaning and analysis:

- Missing platform values could not be reliably reconstructed from the available evidence and were therefore labeled Unknown.

- Missing or unrecoverable text was not fabricated. Unknown was used as an explicit placeholder for unavailable content.
- Strings beginning with NULL were treated as missing placeholders rather than literal post content.
- HTML tags and HTML entities were treated as presentation/encoding artifacts. Tags were removed and entities decoded without intentionally altering the semantic text.
- The targeted Latin-1 → UTF-8 repair was considered appropriate for the 287 suspicious encoding cases because sample conversions were manually/analytically validated before applying the transformation.
- Ten-digit Unix timestamps were interpreted as Unix timestamps in seconds.
- Date-only values in DD-MM-YYYY format were standardized to 00:00:00 after parsing. This is a representation convention and does not imply that the original post was necessarily created exactly at midnight.
- Negative likes were assumed to be invalid count values, rather than positive likes whose signs had been accidentally reversed.
- The median of valid non-negative likes (2,502) was considered an appropriate robust imputation value because likes are bounded count data and may contain extreme observations.
- Pearson correlations were interpreted as measures of linear association only, not evidence of causation.
- Unknown platform records were retained in the EDA so that the original missingness was not hidden

19. Limitations

Despite the systematic cleaning and exploratory analysis, several limitations should be considered when interpreting the results:

- The source data do not provide sufficient external evidence to independently verify the true platform for records with missing platform information.
- The underlying mechanism responsible for the missing values cannot be established from the available data alone.
- Median imputation changes the observed distribution of likes and may affect statistics such as the mean, variance, and correlations involving likes.
- Pearson correlation captures linear association and may fail to detect nonlinear or interaction-based relationships.
- Differences in average engagement across platforms and languages are descriptive comparisons and do not establish causal effects.
- Extreme engagement values were not considered erroneous solely because they were statistically unusual; therefore, legitimate extreme observations may remain in the dataset.
- The Users reference table was used to enrich and validate follower-count information, but follower count should not be interpreted as the sole determinant of engagement.

- The analysis is primarily exploratory and describes patterns present in this dataset; the findings should not automatically be generalized to all social-media platforms, users, or posts.

20. Reproducibility & Code Documentation

The complete data-cleaning and exploratory workflow is reproducible in Python/Google Colab using pandas, NumPy, matplotlib, and standard Python libraries. The analysis was organized into a sequential workflow to ensure that each transformation could be traced, validated, and reproduced.

The execution order is:

Load Data → Baseline Audit → Duplicate Handling → Field-Level Cleaning → Validation → Exploratory Analysis → Export → Reload & Final Verification

The representative documented code used in the cleaning workflow is shown below:

```
# Remove exact duplicate records
```

```
posts_clean = posts.copy()
```

```
posts_clean = posts_clean.drop_duplicates()
```

```
# Treat NULL-prefixed text as missing
```

```
null_mask = (
```

```
    posts_clean["text_content"]
```

```
    .astype(str)
```

```
    .str.strip()
```

```
    .str.contains(r"^NULL", case=False, regex=True, na=False)
```

```
)
```

```
posts_clean.loc>null_mask, "text_content"] = pd.NA
```

```
# Clean HTML and HTML entities
```

```
def clean_text(text):
```

```
    if pd.isna(text):
```

```
        return text
```

```

text = re.sub(r"<[^>]+>", "", str(text))

text = html.unescape(text)

text = re.sub(r"\s+", " ", text).strip()

return text

posts_clean["text_content"] = (
    posts_clean["text_content"].apply(clean_text)
)

# Treat negative likes as invalid
posts_clean.loc[posts_clean["likes"] < 0, "likes"] = pd.NA

# Robust median imputation
median_likes = posts_clean["likes"].dropna().median()

posts_clean["likes"] = (
    posts_clean["likes"]
    .fillna(median_likes)
    .astype("int64")
)

```

Mixed timestamp representations were handled using deterministic parsing rules: 10-digit numeric values were interpreted as Unix timestamps in seconds, DD-MM-YYYY values were parsed using explicit day-first interpretation, and remaining valid representations were processed using the datetime parser.

The final CSV was reloaded after export and independently verified. The verification confirmed that the exported dataset contained 12,000 rows and the original 8 columns, with:

- 0 missing values
- 0 exact duplicate rows
- 0 duplicate post_id values
- 0 negative engagement values

This final reload-and-verify step provides an additional reproducibility check and confirms that the exported file reflects the validated analytical dataset rather than only the in-memory dataframe.

21. Conclusion

The Social Engine Posts dataset was systematically audited, cleaned, validated, and explored using documented, evidence-based, and reproducible procedures. The workflow addressed the major identified data-quality issues while minimizing unnecessary alteration of the underlying observations.

Exact duplicate records were removed, missing platform metadata and unavailable text were represented explicitly using “Unknown” recoverable HTML and encoding artifacts were corrected, and mixed timestamp representations were standardized. Negative likes were identified as semantically invalid and treated as missing before median imputation using the valid non-negative median of 2,502.

Post-cleaning validation confirmed that the final dataset contains 12,000 records across the original 8 columns, with no missing values, duplicate records, duplicate post IDs, or negative engagement values.

The EDA indicates that platform activity is broadly balanced and posting volume remains relatively stable over the 12-month observation period. Likes, shares, and comments exhibit negligible linear correlations, while follower count also shows negligible linear association with all three engagement measures. Platform and language groups display some differences in average engagement, but these differences are descriptive and do not establish causal effects. Similarly, the narrow difference in the proportion of top-10%-like posts across platforms provides no evidence of an overwhelming platform advantage.

Overall, the resulting dataset provides a clean, validated, and reproducible foundation for Phase-1 EDA and subsequent analytical work. The findings should, however, be interpreted within the documented assumptions and limitations, particularly those associated with missing-information treatment, median imputation, categorical comparisons, and the use of Pearson correlation to assess only linear relationships.